PRACTICAL EXERCISE 22

AI Image Generation & Refinement

Exploring textual inversion and layout control by generating a promotional poster for a college coding competition.

PHASE 1: COMPLEX SCENE GENERATION

USER PROMPT

"Create a modern promotional poster for a college-level coding competition. Include a computer screen, programming symbols, students collaborating, a professional layout, and a technology-themed background."



Figure 1: Complex Generation (Abstracted output demonstrating literal interpretation of all prompt elements)

Observations

The initial prompt yields a "kitchen sink" result. The AI includes every requested element, which can create a cluttered composition. The typography competes with the heavy background and character elements.

Prompt Analysis

By listing specific physical objects ("computer screen", "students"), the AI is forced into a literal composition. For a more professional look, prompts must shift from literal objects to aesthetic guidelines.

PHASE 2: MINIMALIST PROMPT REFINEMENT

USER PROMPT

"Generate a minimalist promotional poster for a college coding competition. Use a lot of negative space, clean typography, an elegant white and subtle blue color palette, and abstract geometric representations of code rather than literal objects."



Figure 2: Minimalist Generation (Abstracted output demonstrating constraints and negative space)

KEY LEARNINGS: VISUAL PROMPTING

The refinement phase illustrates a core principle of AI image generation: **less is often more**. When generating professional assets like posters, the AI performs better when directed toward a specific aesthetic style ("minimalist", "swiss design", "flat vector") rather than being overloaded with a list of physical items to include.

By requesting "negative space" and "abstract geometry," we override the AI's tendency to fill every pixel with detail. This exercise proves that successful AI content creation relies on iterative prompting—evaluating the first output, identifying compositional flaws (like clutter), and rewriting the prompt to apply specific stylistic constraints.